

- RAG Learning

2.6 Sequences

Pearson Edexcel GCSE Mathematics 1MA1 — Algebra — spec refs A23–A25 — Papers 1 (non-calculator), 2 and 3 (calculator)

2.6.1 Generating Sequences (A23)

A **sequence** is a list of numbers (**terms**) in a particular order, following a rule. The **position** of a term is written n : the first term is at $n = 1$, the second at $n = 2$, and so on.

Term-to-term rules

A **term-to-term rule** says how to get the next term from the one before. It only works if you also know a starting term.

Worked Example: The first term of a sequence is 4. The term-to-term rule is "multiply by 2 then add 3". Write down the next three terms.

$$4 \times 2 + 3 = 11, \quad 11 \times 2 + 3 = 25, \quad 25 \times 2 + 3 = 53$$

Sequence: 4, **11, 25, 53**

You may see a term-to-term rule written in **suffix notation**, where u_n is the n th term and u_{n+1} is the next one. For example $u_{n+1} = 3u_n - 2$ with $u_1 = 2$ gives 2, 4, 10, 28, ... The question will always explain the notation it uses.

Position-to-term rules (the n th term)

A **position-to-term rule**, or **n th term**, lets you find any term directly by substituting its position for n — no need to list all the terms before it.

Worked Example: The n th term of a sequence is $6n - 1$. Write down the first three terms and work out the 30th term.

$$n = 1: 6 - 1 = 5; \quad n = 2: 12 - 1 = 11; \quad n = 3: 18 - 1 = 17$$

$$n = 30: 6 \times 30 - 1 = \mathbf{179}$$

Worked Example: Write down the first three terms of the sequence with n th term $n^2 - 2$.

$$1 - 2 = -1, \quad 4 - 2 = 2, \quad 9 - 2 = 7 \rightarrow \mathbf{-1, 2, 7}$$

Sequences from patterns and diagrams



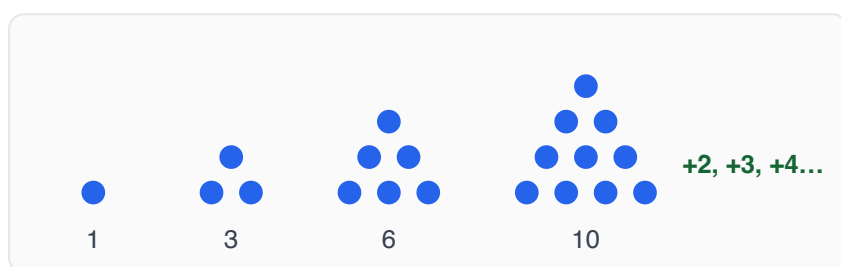
Each extra triangle shares one stick with the previous one, so it needs only 2 more sticks. The numbers of sticks are 3, 5, 7, ... with n th term $2n + 1$, so pattern 50 needs $2 \times 50 + 1 = 101$ sticks.

2.6.2 Special Sequences (A24)

Sequence	First terms	n th term
Square numbers	1, 4, 9, 16, 25, 36, ...	n^2
Cube numbers	1, 8, 27, 64, 125, ...	n^3
Triangular numbers	1, 3, 6, 10, 15, 21, ...	$\frac{n(n+1)}{2}$

Even numbers	2, 4, 6, 8, ...	$2n$
Odd numbers	1, 3, 5, 7, ...	$2n - 1$
Powers of 3	3, 9, 27, 81, ...	3^n

Triangular numbers



Each new triangle of dots adds a row that is one dot longer than the last, so the gaps between triangular numbers are 2, 3, 4, 5, ... Using the formula, the 15th triangular number is $\frac{15 \times 16}{2} = 120$.

Arithmetic progressions

In an **arithmetic (linear)** sequence the same number is added each time. This is the **common difference** d , which can be negative: 25, 21, 17, 13, ... has $d = -4$.

Fibonacci-type sequences

In a **Fibonacci-type** sequence each term (after the first two) is the **sum of the two terms before it**. The Fibonacci sequence is 1, 1, 2, 3, 5, 8, 13, 21, ... but the rule can start from any two numbers.

With first two terms a and b , the terms are a , b , $a + b$, $a + 2b$, $2a + 3b$, $3a + 5b$, ... Writing terms like this turns "find the first two terms" questions into simultaneous equations.

Worked Example: The 3rd term of a Fibonacci-type sequence is 9 and the 5th term is 23. Work out the first two terms.

$$\text{3rd term: } a + b = 9 \quad \text{5th term: } 2a + 3b = 23$$

Double the first: $2a + 2b = 18$. Subtract from the second: $b = 5$, so $a = 4$.

Sequence: **4, 5**, 9, 14, 23 ✓

Geometric progressions

In a **geometric** sequence each term is **multiplied** by the same number, the **common ratio** r . Find r by dividing any term by the term before it. On the Foundation tier r is a positive rational number, such as 4, $\frac{1}{3}$ or 1.5.

Geometric sequence with first term a and common ratio r :

$$a, ar, ar^2, ar^3, \dots \quad n\text{th term} = ar^{n-1}$$

Worked Example: A geometric sequence has first term 3 and a positive common ratio. The 3rd term is 48. Work out the 5th term.

$$3r^2 = 48 \Rightarrow r^2 = 16 \Rightarrow r = 4$$

Terms: 3, 12, 48, 192, **768**

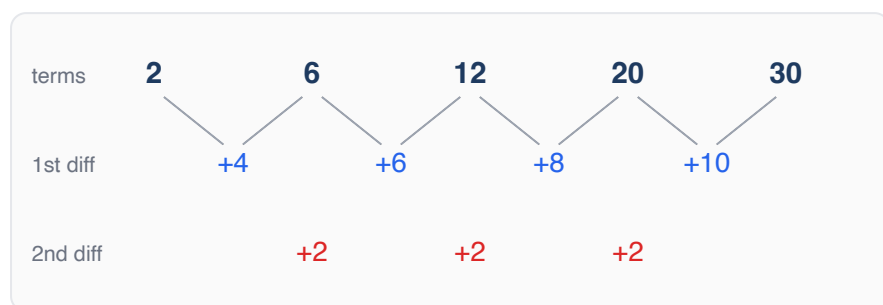
Worked Example: A geometric sequence begins 81, 27, ... Write down the next two terms.

$$r = 27 \div 81 = \frac{1}{3}, \text{ so the next terms are } 27 \times \frac{1}{3} = \mathbf{9} \text{ and } \mathbf{3}.$$

Quadratic sequences

In a **quadratic** sequence the first differences are not constant, but the **second differences** (the differences between the differences) are. Recognising and continuing quadratic sequences is on

both tiers.



The second difference is always 2, so 2, 6, 12, 20, 30, ... is a quadratic sequence. The next first differences are +12 and +14, giving the next terms 42 and 56.

Exam Tip: Identify the type first: same first difference $\Rightarrow$ arithmetic; same second difference $\Rightarrow$ quadratic; same ratio $\Rightarrow$ geometric; each term is the sum of the two before $\Rightarrow$ Fibonacci-type.

Surd ratios and other sequences Higher

On the Higher tier the common ratio of a geometric sequence can be a **surd**, and you can be asked about "other sequences" such as ones with a negative ratio, or rules defined in the question.

Worked Example: A geometric sequence begins 5, $5\sqrt{3}$, 15, ... Work out the 7th term.

$$r = \frac{5\sqrt{3}}{5} = \sqrt{3}$$

$$\text{7th term} = 5 \times (\sqrt{3})^6 = 5 \times 3^3 = 5 \times 27 = \mathbf{135}$$

Worked Example: A geometric sequence has first term 4 and second term $4\sqrt{2}$. Find the first term that is greater than 1000.

$$r = \sqrt{2}, \text{ so every two terms the value doubles: } 4, 4\sqrt{2}, 8, 8\sqrt{2}, 16, \dots$$

$$\text{The odd-numbered terms are powers of 2 times 4: } 17\text{th term} = 4 \times 2^8 = 1024.$$

16th term $= 4 \times 2^7 \times \sqrt{2} = 512\sqrt{2} \approx 724.1$, which is less than 1000. So the answer is the **17th** term (1024).

A negative common ratio makes the terms alternate in sign: 3, -6 , 12, -24 , 48, ... has $r = -2$.

2.6.3 The n th Term of a Linear Sequence (A25)

For a linear sequence with common difference d :

$$n\text{th term} = dn + (\text{first term} - d)$$

"First term $-d$ " is the **zeroth term**: the number that would come just before the first term.

Worked Example: Find the n th term of 5, 12, 19, 26, ...

The common difference is 7, so compare with $7n$: 7, 14, 21, 28, ...

Each term is 2 less than $7n$, so the n th term is **$7n - 2$** .

Worked Example: Find the n th term of 50, 44, 38, 32, ... and work out the 20th term.

$d = -6$; zeroth term $= 50 + 6 = 56$. n th term $= \mathbf{56 - 6n}$

20th term $= 56 - 120 = \mathbf{-64}$

Is a number in the sequence?

Set the n th term equal to the number and solve for n . The number is a term only if n is a **positive integer**.

Worked Example: Is 131 a term of $9n - 4$? $9n - 4 = 131 \Rightarrow 9n = 135 \Rightarrow n = 15$. Yes, it is the 15th term.

Is 90 a term of $5n + 2$? $5n = 88 \Rightarrow n = 17.6$. This is not a whole number, so **no**.

Problems with linear sequences

Worked Example: Sequence A has n th term $6n + 1$. Sequence B begins 79, 76, 73, ... Find the position at which the two sequences have the same term.

B: $d = -3$, zeroth term 82, so n th term $= 82 - 3n$.

$6n + 1 = 82 - 3n \Rightarrow 9n = 81 \Rightarrow n = 9$. Both 9th terms are **55**.

Worked Example: 1 table seats 5 people, 2 tables pushed together seat 8, and 3 tables seat 11. What is the least number of tables needed to seat 60 people?

n th term $3n + 2$. Need $3n + 2 \geq 60 \Rightarrow n \geq 19.33 \dots$, so **20** tables (19 tables seat only 59).

Exam Tip: The number in front of n is the common difference, not the first term — 4, 7, 10, 13 is $3n + 1$, not " $n + 3$ ". Check your expression by substituting $n = 1$ and $n = 2$ before moving on.

2.6.4 The n th Term of a Quadratic Sequence Higher (A25)

For a quadratic sequence with n th term $an^2 + bn + c$:

$$a = \frac{\text{second difference}}{2}$$

1. Work out the second difference and halve it to get a .
2. Subtract an^2 from each term of the sequence.
3. Find the n th term of what is left (a linear sequence $bn + c$).
4. Combine to give $an^2 + bn + c$, and check with $n = 1$ and $n = 2$.

Worked Example: Find the n th term of 4, 11, 22, 37, 56, ...

First differences: 7, 11, 15, 19 Second difference: 4, so $a = 2$.

n	1	2	3	4	5
Term	4	11	22	37	56
$2n^2$	2	8	18	32	50
Term $- 2n^2$	2	3	4	5	6

What is left is $n + 1$, so the n th term is $2n^2 + n + 1$. Check: $n = 3$: $18 + 3 + 1 = 22 \checkmark$

Worked Example: Find the n th term of 10, 15, 18, 19, 18, ...

First differences: 5, 3, 1, -1 Second difference: -2 , so $a = -1$.

Term $-(-n^2)$, i.e. term $+n^2$: 11, 19, 27, 35, 43 $\Rightarrow 8n + 3$

n th term $= -n^2 + 8n + 3$. Check: $n = 2$: $-4 + 16 + 3 = 15 \checkmark$

Worked Example: The n th term of a sequence is $n^2 + 3n$. Show that 130 is a term of the sequence.

$$n^2 + 3n = 130 \Rightarrow n^2 + 3n - 130 = 0 \Rightarrow (n + 13)(n - 10) = 0$$

$n = 10$ (reject $n = -13$, since positions are positive). 130 is the **10th** term.

Exam Tip: The coefficient of n^2 is *half* the second difference — using the full second difference is the most common error. A negative second difference gives a negative n^2 term.

Quick Reference Summary

Sequence type	How to spot it	n th term	Tier (1MA1)
Arithmetic (linear)	Constant first difference d	$dn + (\text{first term} - d)$	Foundation
Geometric	Constant ratio r (term $\div$ previous term)	ar^{n-1}	Foundation (r positive rational); Higher (r a surd, or other sequences)
Quadratic	Constant second difference	$an^2 + bn + c$, $a = \text{half the second difference}$	Recognise and continue: Foundation; n th term: Higher
Fibonacci-type	Each term = sum of the previous two	Use $a, b, a + b, a + 2b, \dots$	Foundation
Square / cube / triangular	1, 4, 9... / 1, 8, 27... / 1, 3, 6...	$n^2 / n^3 / \frac{n(n+1)}{2}$	Foundation